

 **Tip:** The Bootstrap framework is open source and the source code of the framework is available on GitHub. Developers/programmers are encouraged to participate and contribute to the project.

Express.js framework

Express.js or Express is a Web application framework for Node.js designed for building Web applications and APIs. It is the *de facto* standard server framework for Node.js. Express is also the back-end part of the MEAN stack, together with the MongoDB database and the AngularJS front-end framework.

Some of the salient features of the Express.js framework are:

- It's a cross-platform framework, which means it is not limited to one OS.
- It is a server-side Web and mobile application framework written in JavaScript.
- It provides Express Generator, which allows you to create complex applications quickly.
- It supports the MVC pattern.
- Express.js comes with two template engines, Jade and EJS, which facilitate the flow of data into the structure of a website.
- It has a provision for building single-page, multi-page and hybrid Web and mobile applications, as well as APIs or Application Programming Interfaces.

Installing Node.js

Express.js is a Node.js based framework, so Node.js must be installed first. Download and install Node.js (<https://nodejs.org/>). Express.js is distributed as an *npm* package, so we will use Node and *npm* to install Express.js.

Building a website

Let us build a website using these two frameworks – Bootstrap and Express.js.

We will use open source programming languages like HTML, CSS and JavaScript along with these two frameworks, to create a website.

 **Note:** It is assumed that you have a basic knowledge of Web technologies like HTML, CSS and JavaScript. If you don't, W3Schools (<http://www.w3schools.com/>) is a good place to start. The site has some great tutorials for Web technologies which are easy to follow.

First, let's create a folder/directory for our website.

Open the command prompt/terminal and type the following command:

```
C:>mkdir project
```

This command creates a folder called *project*:

```
C:>cd project
```

This command will make *project* your working directory.

Now, use the *npm init* command to create a very important *package.json* file for the website/ application. Run the following command in the newly created folder, i.e., the *project* folder:

```
C:\project>npm init
```

This command prompts you to enter certain fields, such as name, version of your website or application. You can either hit the *Enter* key and accept the default setting or enter the details you want, with the following exception:

Entry point: (*index.js*)

At this point, you will be asked to enter the entry point file name. If you hit *Enter*, it will set the *index.js* file as the entry point, by default. But for this tutorial, we are going to give it a new name. Let's name it *website.js*. Type *website.js* and press the *Enter* key.

Now, let us install Express in the *project* directory and save it in the dependencies list. Run the following command in the *project* folder:

```
C:\project>npm install express -save
```

The above command will install all the important Express modules, and save the dependencies in the *package.json* file.

The *package.json* file will look like what follows:

```
{
  "name": "project",
  "version": "1.0.0",
  "description": "Sample Project",
  "main": "website.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "author": "AniketKudale",
  "license": "ISC"
}
```

Here's a description of the code snippet given above.

- *name*: Name of the Web application.
 - *version*: Version of the Web application.
 - *description*: Description of the Web application.
 - *main*: Main entry point of the Web application; in this file, we store our application logic.
 - *scripts*: Here, we specify script commands that run at various intervals in the life cycle of our package.
 - *author*: Here, we specify the author's name, i.e., your name.
 - *license*: Here, we can specify the licence type.
- The Express.js framework has now been successfully